



CY 104 – Intro to Cybersecurity

Assignment # 01

Control structures

—

Student Name: Athar Imran

Student ID: 2604277

Assignment Date: 6th October, 2026

Assignment done: 6th October, 2026

Due Date: 6th October, 2026

Incidents table

NO.	THREAT IDENTIFIED	ASSET AT RISK	THREAT ACTOR	ATTACK VECTOR / METHOD	CIA PROPERTY AFFECTED	AAA ISSUE	EXTERNAL / INSIDER
2	Brute-force & password guessing	Staff accounts & university records	External attacker	Repeated unsuccessful login attempts	**Confidentiality, Availability**	**Authentication**	**External**
3	Unauthorized access & account misuse	Student information	Insider or employee	Using another employee's account	**Confidentiality, Integrity**	**Authentication, Authorization**	**Insider**
4	Data theft & unauthorized copying	Student's personal records	Insider or staff member	Copying records to personal USB	**Confidentiality**	**Authorization, Accounting**	**Insider**
5	Weak authentication & password sharing	User accounts and academic records	Insider or user	Simple passwords and sharing passwords	**Confidentiality, Integrity**	**Authentication**	**Insider**
6	Pretexting & social engineering	Employee login credentials	External attacker	Pretending to IT support on phone	**Confidentiality**	**Authentication**	**External**
7	Malware & malicious attachment	Administration computer and data	External attacker or malware creator	Unknown email attachment	**Confidentiality, Integrity, Availability**	**Authentication**	**External**
8	Lack of logging & accountability	Sensitive information and records	Insider or external attacker	Unauthorized access/modification without proper logs	**Confidentiality, Integrity**	**Accounting**	**Both**

AAA weaknesses

- **Authentication** weaknesses allows attackers & unauthorized users to gain or attempt access, as seen in **Incidents 2, 3, 5 and 6**.
- **Authorization** weaknesses allows an employee to access other employee's account & confidential records as in **Incidents 3 and 4**.
- **Accounting** weaknesses makes it challenging to track who accessed or copied information, as in **Incidents 4 and 8**.

Insider threats

Incidents 3, 4 and 5.

Insider threats are harder to detect as because employees have legitimate access to university systems and information, already. Their activities look similar to authorized work, making unauthorized actions difficult to visible without proper logging & monitoring.